Maths

Unit 3 – Averages

22.10.26

WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.

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I can identify what the mean, mode and median is.



I can find the mean, mode and median from a set of data.



I can solve word problems involving the mean, mode and median.

WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.

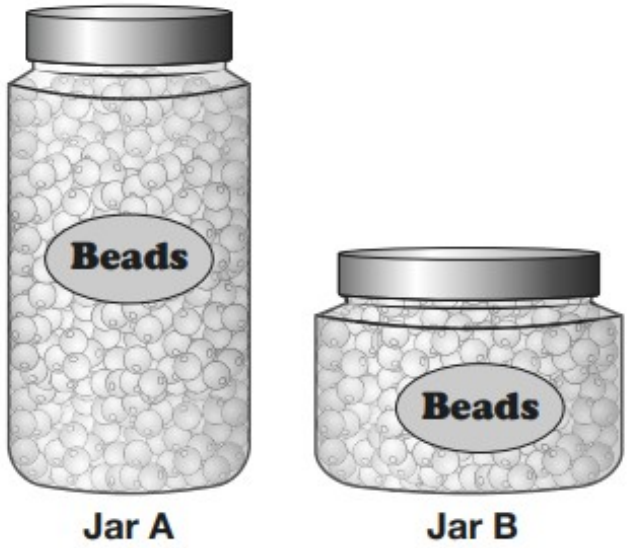
Retrieval Practice

Explain why Olivia is correct?

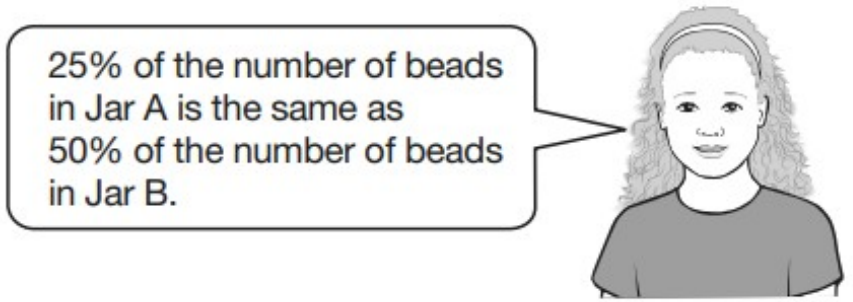
She is correct because Jar A is double Jar B, 25% doubled is 50% which means that if Jar A had 100 beads, 25% would be 25 beads and 50% of Jar B would be 25.

Olivia has two jars of beads.

The number of beads in Jar A is **double** the number of beads in Jar B.



Olivia says,



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What do you know?

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We are going to investigate the mean, mode and median of our shoe sizes in our groups.

You have a post it note.

On your post it note, I want you to write your shoe size.

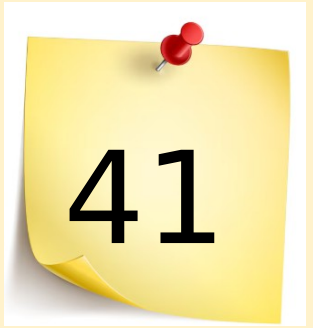














In your group, you need to order your shoe sizes from smallest to largest.

If there is a more than one of a size, you need to put it on top of the other.













Can anyone see a pattern with the data?

What do multiple of the same number show?**The mode**

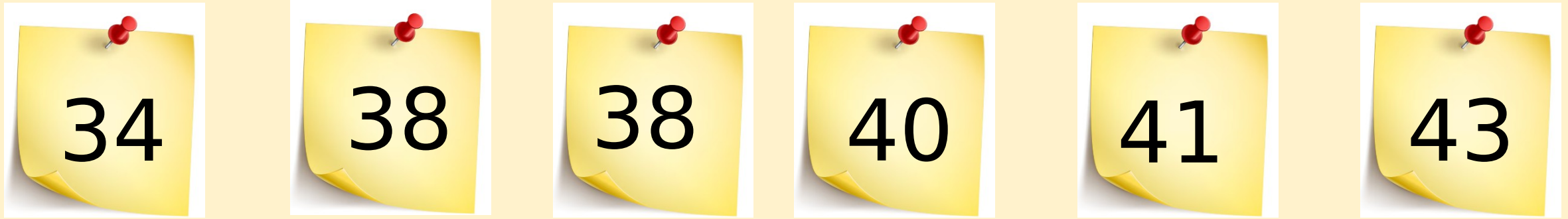
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Mode

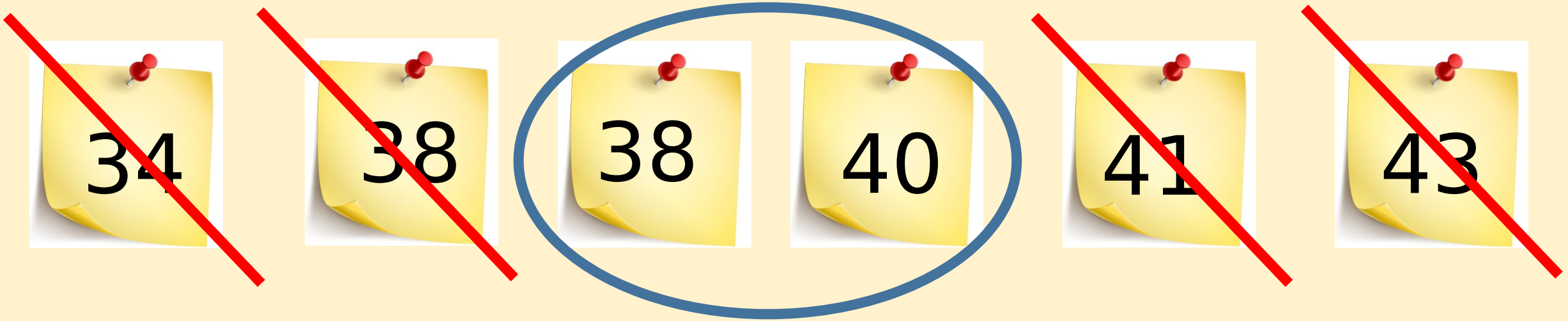
The mode of a set of data is the number that appears the most.

With mode, there can be one, more than one or none!

Did you know, if there is 2 modes, it is know as **bi-modal.**



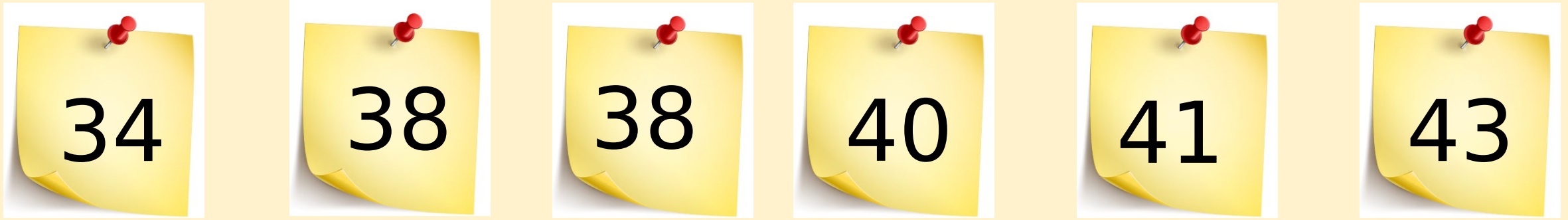
Now, you need to order your data from smallest to largest in a straight line.



I want you to find the median from your set of data.

How do you think we do this?

Find the value that is in the middle of the set of data once it has been ordered from smallest to largest.



$38 + 40 = 78$

$78 \div 2 = 39$

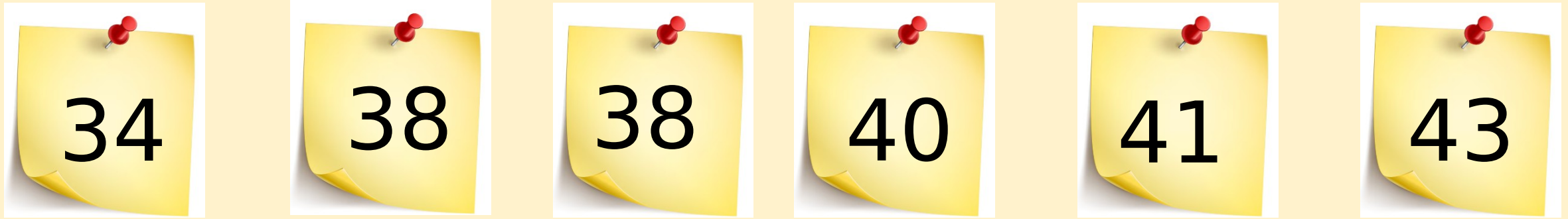
You now to calculate the **median** of your set of data.

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Median

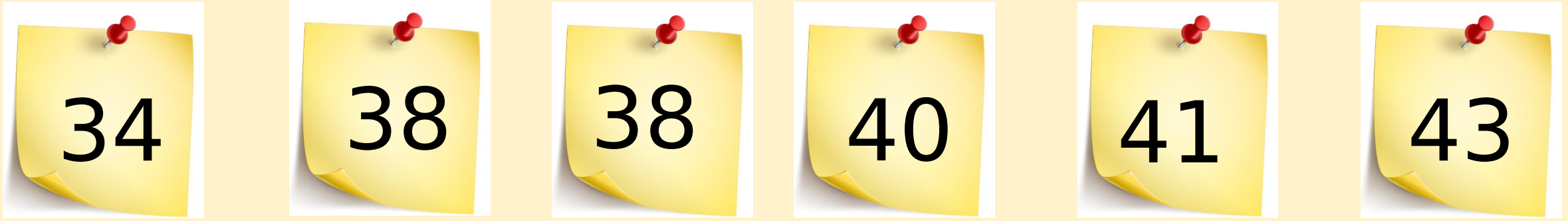
The median is the value that is in the middle of the data.

If there are two values, we need to add the two values and then divide by 2.



The final average we are going to find is the mean.

To find the mean, we need to add all the values together and then divide by the total number of values.



$34 + 38 + 38 + 40 + 41 + 43 = \mathbf{234}$

$234 \div 6 = \mathbf{39}$

You now need to calculate the **mean** of your set of data.

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Mean

The mean is the value that is achieved when you add all the values and divide them by the total number of numbers.

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The mode, mean and the median are the 3 types of averages.

We can also find the **range**. The range shows us the how spread out the data is.

For this, we take the largest value and subtract the smallest value.



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WALT: Understand that the mode, median, mean and range are ways to describe and summarise data sets.

In your notebooks, write a definition for:



Mode –

Median –

Mean –

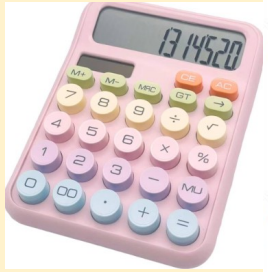
Finished?

Can you explain to your partner how to find each type of average?

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Your task





Given	Mean	Median	Mode
a. 10, 12, 10, 17, 11			
b. 6, 8, 11, 5, 4, 8			
c. 34, 38, 38, 22, 24			
d. 43, 32, 31, 43, 21, 23, 17			
e. 54, 64, 65, 73, 64			
f. 7, 8, 9, 10, 12, 13, 11, 10, 5			



Finished?

Take a challenge and answer it in your book.
Once you've finished one, get another one.

PROBLEM 1:

Chef Maya manages the Sweet Treats Bakery. She wants to stock the right amount of ingredients for next week, so she records the total number of cupcakes sold each day over a 7-day week.

Cupcakes sold each day during the week: 24, 30, 18, 24, 35, 24, 27

1. Order Data	Least to greatest: [_____ , _____ , _____ , _____ , _____ , _____ , _____]
2. Find the Mode	Mode = _____
3. Find the Median	Median = _____
4. Find the Mean	Mean = _____
5. Find the Range	Range = _____ - _____ = _____

If Chef Maya wants to stock ingredients based on her most frequent daily sales level, which measure (Mean, Median, or Mode) should she look at? Why?

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Given	Mean	Median	Mode
a. 10, 12, 10, 17, 11 10, 10, 11, 12, 17 10 + 10 + 11 + 12 + 17 = 60 60 ÷ 5 = 12	12	11	10
b. 6, 8, 11, 5, 4, 8 4, 5, 6, 8, 8, 11 4 + 5 + 6 + 8 + 8 + 11 = 42 42 ÷ 6 = 7	7	(6+8)÷2 =7	8
c. 34, 38, 38, 22, 24 22, 24, 34, 38, 38 22 + 24 + 34 + 38 + 38 = 156 156 ÷ 5 = 31.2	31.2	34	38
d. 43, 32, 31, 43, 21, 23, 17 17, 21, 23, 31, 32, 43, 43 17 + 21 + 23 + 31 + 32 + 43 + 43 = 210 210 ÷ 7 = 30	30	31	43
e. 54, 64, 65, 73, 64 54, 64, 64, 65, 73 54 + 64 + 64 + 65 + 73 = 320 320 ÷ 5 = 64	64	64	64
f. 7, 8, 9, 10, 12, 13, 11, 10, 5 5, 7, 8, 9, 10, 10, 11, 12, 13 5 + 7 + 8 + 9 + 10 + 10 + 11 + 12 + 13 = 85 85 ÷ 9 = 9.4	9.4	10	10

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Plenary -

Mr. Jon says,

“I have found the mode for a set of data. The mode is 0 which means the data has no mode.”

Is he correct?

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Homework

ESL – Workbook 3.1 + 3.2

Maths – Workbook 37-42

Due tomorrow!